

Amir Hossein Amanzadi

DOCTORAL RESEARCH FELLOW

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Education

University of Oslo

PH.D. IN MEDICINE AND HEALTH SCIENCES

- Supervisors: Prof. Ole Andreassen, Dr. Dennis van der Meer

Oslo, Norway

Jan 2026 - Present

Uppsala University

M.SC. IN PHARMACEUTICAL SCIENCE - DRUG DISCOVERY & DEVELOPMENT

- Supervisors: Prof. Ola Spjuth, Dr. Narsis Kiani

Uppsala, Sweden

Sep 2019 - Jul 2021

Sharif University of Technology

B.SC. IN PURE CHEMISTRY & MECHANICAL ENGINEERING (MINOR)

- Supervisors: Prof. Ali Pourjavadi, Prof. Amir Shamloo

Tehran, IRAN

Mar 2013 - Aug 2018

Work Experience

Oslo University Hospital

DOCTORAL RESEARCH FELLOW

- Engineered advanced agentic AI workflows and local AgentOps infrastructure for precision psychiatry, integrating multi-omics, imaging, and metabolomics data to accelerate clinical risk prediction iterations 2x.
- Spearheaded Metabolome-Wide Association Studies (MWAS) investigating the genetic and metabolic overlap between brain-related diseases and cardiometabolic traits.
- Enforced strict data governance and AI ethics protocols across secure, air-gapped platforms to safeguard sensitive, large-scale biobank datasets.

Oslo, Norway

Jan 2026 - Present

Diverge AI (Stealth Startup)

AI SOLUTION ARCHITECT

- Founded Medhoush, a clinically validated multimodal AI chat platform, serving 400,000 active users across major social media platforms, including Telegram and WhatsApp.
- Founded Yara, an AI-powered (VLM) radiology assistant designed to streamline diagnostic workflows in healthcare systems, enhancing efficiency and accuracy in medical imaging analysis.
- Built end-to-end LLMops tools (fine-tuning/RAG/local inference), cutting costs by 30% and accelerating model iterations 2x.

Dubai, UAE

Nov 2024 - Dec 2025

Prepaire Labs

HEAD OF BIOINFORMATICS

- Established Hailo, an innovative drug discovery platform that integrates large language models (LLMs) with robotic wet lab systems, enhancing drug development screening and pandemic response capabilities.
- Developed Clicktromics, an advanced generative AI framework for immunotherapy, aimed at optimizing the design of antibody-drug conjugates.
- Led a team of 12 interdisciplinary researchers in designing and developing cutting-edge AI tools to expedite the advancement of adoptive cell therapies.

Abu Dhabi, UAE

Sep 2023 - Oct 2024

Karolinska University Hospital

CLINICAL DATA SCIENTIST

- Collaborated closely with Prof. Jesper Tegnér and Prof. Pierre-Emmanuel Rautou, MD within the framework of the DECISION EU Project. Our focus was the analysis of three cohorts of over 2500 decompensated cirrhosis patients. We successfully identified three highly promising combinatorial treatments. Presently, undergoing phase II clinical trials.

Stockholm, Sweden

Mar 2022 - Jun 2023

Celeris Therapeutics

HEAD OF COMPUTATIONAL CHEMISTRY

- Led a team of three computational chemists and established Xanthos, an active learning PROTAC and Molecular Glue design engine.
- Developed a Geometric deep learning method for predicting Protein-Protein Interaction susceptible to Targeted Protein Degradation.
- Have identified and patented hits via in-silico screening against alpha-synuclein (Parkinson's disease) that is currently in Lead optimization
- Contributed directly to advancing key partnership projects with Merck kGaA, Boehringer Ingelheim, and IRBM.

Menlo Park, CA, USA

Apr 2021 - Mar 2022

Karolinska Institute

MACHINE LEARNING RESEARCHER

- Collaborated with Dr. Hassan Abolhassani, MD on developing a robust few-shot learning method for early detection of Ataxia-Telangiectasia disorder in children based on their genetic profiles.

Stockholm, Sweden

Aug 2020 - Jan 2021

Shenakht Pajouh (Cognition Research)

CO-FOUNDER & MACHINE LEARNING ENGINEER

- Contributed to the development of various generative models used to establish a conversational AI agent for people in mental health crises.
- Worked with Dr. Reza Lashgari on designing an AI-based EEG interface for binary communication with Coma patients.

Tehran, Iran

Jan 2017 - Mar 2019

Research Experience

Uppsala University

GUEST LECTURER

Uppsala, Sweden

Sep 2022 - Sep 2024

- Teach the application of Graph Neural Networks (GNN) on Biomedical Research for the Nationwide Artificial Intelligence in Drug Discovery remote course in Sweden.

European Foundation for the Study of Chronic Liver Failure (EF CLIF)

ASSOCIATE RESEARCHER

Barcelona, Spain

Oct 2020 - Feb 2022

- Facilitated Dr. David Gomez-Cabrero by developing a multimodal ML model for the prognosis of Chronic Liver Failure from the multi-omics data

Center for Molecular Medicine (CMM)

MASTER THESIS INTERN

Stockholm, Sweden

Jan 2021 - Jun 2021

- Under the supervision of Prof. Ola Spjuth & Dr. Narsis Kiani, I established a chemistry-informed and interpretable Graph Convolutional Network (GCN) model that predicts polypharmacy side effects of drug combinations based on their SMILES.

Algorithmic Dynamics Lab

SUMMER RESEARCH INTERN

Solna, Sweden

Jun 2020 - Aug 2020

- Assisted Dr. Hector Zenil in investigating the applicability of Algorithmic Information Dynamics in the causal discovery of synergistic mechanisms behind drug combinations. The project was part of the AstraZeneca-Sanger Drug Combination DREAM Consortium.

Karolinska Institute

GRADUATE RESEARCH ASSISTANT

Stockholm, Sweden

Nov 2019 - Jun 2020

- Aided Dr. Narsis Kiani during the pandemic, by developing a variety of graph representation learning models to identify clinically viable drug combinations for diabetic patients suffering from COVID-19 comorbidity. Project was funded by the Novo Nordisk Foundation.

Sharif University of Technology

RESEARCH ASSISTANT

Stockholm, Sweden

Nov 2019 - Jun 2020

- Designed multifunctional peptides for Alzheimer's disease. Led by Prof. Amir Shamloo.
- Synthesised a green bio-compatible hydrogel for rapid wound healing, under the supervision of Prof. Ali Pourjavadi.

Teaching Experience

2022-24 **Artificial Intelligence in Drug Discovery**, Guest lecturer

Uppsala University

2017 **Advance Molecular Dynamics Simulation**, Teaching assistant

Sharif Uni.

Honors & Grants

2025 **Full Research scholarship**, Mila - Quebec AI Institute

Canada

2025 **Exemption scholarship**, University of Montréal

Canada

2022 **Research Sponsorship**, European Foundation for the Study of Chronic Liver Failure (EF Clif)

Spain

2021 **First Rank Student**, Pharmaceutical Science master's program in Uppsala University

Sweden

Publications

Protein-Protein Interaction Prediction for Targeted Protein Degradation

OLIVER ORASCH, NOAH WEBER, MICHAEL MÜLLER, **AMIR H. AMANZADI**, CHIARA GASBARRI, CHRISTOPHER TRUMMER

2022

International Journal of Molecular Sciences 23.13 (2022). 2022

Predicting safe drug combinations with Graph Neural Networks (GNN) [Thesis]

AMIR H. AMANZADI

2021

Uppsala University (2021). 2021

Designing a new multifunctional peptide for metal chelation and A β inhibition

AMIR SHAMLOO, MOHSEN ASADBEGI, VAHID KHANDAN, **AMIR H. AMANZADI**

2020

Polymer Engineering & Science 60.5 (2020) pp. 889–899. 2020

Both Tough and Soft Double Network Hydrogel Nanocomposite Based on O-Carboxymethyl Chitosan/Poly(vinyl alcohol) and Graphene Oxide: A Promising Alternative for Tissue Engineering

ALI POURJAVADI, ZAHRA MAZAHARI TEHRANI, HAMID SALAMI, FARZAD SEIDI, ANAHITA MOTAMEDJ, **AMIR H. AMANZADI**,

2020

EHSAN ZAYERZADEH, MEISAM SHABANIAN

Polymer Engineering & Science 60.5 (2020) pp. 889–899. 2020

Skills

Cheminformatics	RDKit, Schrödinger, GROMACS, Gaussian, AutoDock, OpenMM
Programming	Python, R, Elixir, Julia, MATLAB, Linux, Shell, LaTeX
GenAI & LLMs	LangChain/Graph, Pydantic AI, Agno, Dify, Ollama, HuggingFace
ML and Data	PyTorch, TensorFlow, PyG, DGL, Scikit-learn, Numpy, Pandas, Spark, SQL
Soft Skills	Leadership, Communication, Teamwork, Adaptability, Problem-solving
DevOps	AWS, Azure, Docker, kubernetes, Git, Cloudflare, Jira, tmux